

VARUN CHOUDHARY

[✉ cvarun4545@gmail.com](mailto:cvarun4545@gmail.com) | [📞 408-592-9477](tel:408-592-9477) | [📍 Davis, CA](https://www.davis.ca) | [🌐 Portfolio](#) | [🐙 github.com/vcro45](https://github.com/vcro45) | [🌐 linkedin.com/in/varunc4545](https://www.linkedin.com/in/varunc4545)

EDUCATION

B.S. Computer Science

Expected June 2027

University of California, Davis · Davis, CA Coursework: Data Structures & Algorithms Design, ML, DBS, Computer Architecture, OS

EXPERIENCE

Co-Founder & Lead Engineer

2023 – 2025

Lurnigo · San Francisco, CA / Remote *EdTech platform for college admissions — 3,000+ paid students across USA & Australia*

- Identified a gap in the college admissions market, pitched the product concept, and secured full operational backing; drove all product and engineering decisions from zero to a live platform serving **3,000+ paid students** across the USA and Australia within 2 years.
- Designed and built the core **Probability Engine** — a React/Node.js system ingesting student inputs (GPA, test scores, activities, essays) to compute real-time acceptance likelihood, the platform's primary differentiator and top retention driver.
- Owned the product roadmap across **bi-weekly board meetings**, deciding feature launch sequencing across SAT/ACT prep, counsellor matching, and university discovery modules based on student feedback and usage data.
- Automated the mentor payment pipeline (Notion → Stripe), eliminating manual ops overhead and enabling the platform to scale to 8 team members without additional administrative headcount.

Software Engineering Intern

Summer 2024

Leap AI · Santa Clara, CA *Cross-platform AI Chatbot — Web, Android & iOS*

- Led a 6-person engineering team to design, build, and ship a **full-stack AI chatbot** across Web, Android, and iOS in 4 weeks — from schema design to a fully demoed, documented product.
- Built the backend using **FastAPI + PostgreSQL** with chat history persistence and session management; integrated **OpenAI GPT-4** for context-aware responses and Redis caching to reduce latency on repeat queries.
- Architected **Server-Sent Events (SSE)** for real-time streaming on the web client; built native mobile UIs in **Kotlin/Jetpack Compose** (Android) and **SwiftUI** (iOS), all consuming a shared REST API.
- Delivered all platform targets on schedule with bi-weekly check-ins, API documentation, and QA coverage across error, offline, and load states.

PROJECTS

IPLAuction — Real-Time Multiplayer Auction Platform [🔗 live](#)



Node.js · Socket.IO · Next.js · PostgreSQL · Drizzle ORM · Zustand · Railway · Vercel

- Built a multiplayer IPL-style auction platform supporting up to 10 simultaneous participants — users join via room code, select franchises, and bid against each other in real time with a fixed purse and squad constraints.
- Engineered a **serialized bid processing queue** to eliminate race conditions between concurrent bids and timer expirations; strict validation enforces purse limits, squad caps, overseas player rules, and bid increment logic on every submission.
- Implemented an **absolute-timestamp timer model** for accurate countdown sync across all clients, with automatic reset on valid bids and sub-100ms state propagation via Socket.IO.
- Designed a **hybrid state architecture** — auction state held in-memory for performance with selective PostgreSQL checkpointing for bids, lot results, and purse updates; handles reconnects, duplicate tabs, and disconnections via a grace-period session model.
- Shipped a broadcast-style auction UI — live bid feeds, purse tracking, squad view, host controls, and real-time GIF chat.

Pro Ro — UC Davis Student Platform [🔗 live](#)



Next.js · TypeScript · PostgreSQL · Drizzle ORM · NextAuth · GitHub Actions · Vercel

- Built a **UC Davis-exclusive** full-stack platform consolidating course discovery, professor reviews, a campus discussion board, and lost & found — replacing fragmented tools like Reddit and Discord for verified students.
- Implemented a **privacy-first identity layer**: Google OAuth restricted to @ucdavis.edu emails, public username system with DB-enforced uniqueness, 14-day change cooldown, and zero exposure of real identities in public interactions.
- Designed a relational schema with UUID primary keys, composite unique indexes, and FK cascade constraints across 8+ entities; managed evolution via **Drizzle migrations** synced between Docker (local) and Neon (production).
- Built a **CI pipeline** (lint → typecheck → build) required before every merge; V1 is live in production with test users ahead of campus launch.

MockMate — AI Voice Interview Coach [🔗 live](#)



OpenAI Realtime API · WebSockets · React · Node.js

- Built a real-time AI interview coach using **OpenAI's Realtime API** for voice-to-voice interaction with sub-second response latency, simulating realistic recruiter timing and conversational pressure.
- Implemented **bi-directional WebSocket** audio streaming with full-duplex state management, handling natural interruption and turn-taking between the user and AI without lag or dropped state.
- Engineered a post-session feedback engine that evaluates filler word frequency, tonal confidence, and answer keyword relevance — surfacing results on a granular **React scorecard dashboard**.

SKILLS

Languages

Python, JavaScript/TypeScript, Java, C++, SQL, Kotlin, Swift

AI & ML

OpenAI API, PyTorch, HuggingFace, LangChain, DistilBERT, NLP, Scikit-learn

Backend & Infra

Node.js, FastAPI, PostgreSQL, Redis, Docker, Socket.IO, WebSockets, SSE, AWS, CI/CD

Frontend

React, Next.js, Tailwind CSS, Zustand, SwiftUI, Jetpack Compose

Auth & Security

NextAuth, JWT, RBAC, OAuth 2.0, OWASP API Security